Event-Driven and Many-Threaded Architectures for High-Frequency Trading

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***Abstract*—An FPGA can arbitrate NASDAQ market data in six nanoseconds. A spiking neural network trained without supervision can predict price spikes and return 76.8% in simulated trading over nineteen days. These two results sit at opposite ends of a research intersection that few research papers attempt to connect. This survey synthesises nineteen papers spanning FPGA hardware acceleration, event-driven computing, neuromorphic chip architecture, and spiking neural network approaches to high-frequency trading. We classify each paper by methodology and evidence strength and surface assumptions the field shares but does not test. The central proven claim is that purpose-built hardware accelerates financial workloads by orders of magnitude over software. The central gap is that no fabricated neuromorphic chip has been measured on a realistic trading task, and no study has modelled what happens when architecturally heterogeneous trading systems enter the markets they trade in.**

***Index Terms*—FPGA, high-frequency trading, neuromorphic computing, spiking neural networks, event-driven architecture, hardware acceleration.**

I. Introduction

In 2013, Jump Trading purchased a microwave tower in Belgium because signals travel roughly 30-50% faster through air than they do through a fibre-optic cable, allowing them to receive price updates from London and send orders to Frankfurt faster than their competitors using traditional underground cables. Proprietary trading shops use co-location; the practice of placing their electronic devices physically inside the exchange's data centre, such that the market data signals travel the shortest possible distance. In the field of High-Frequency Trading (HFT), being second is often synonymous to being last; a delay of a few nanoseconds translates to an order arriving after the price has already changed, leading to "slippage" or missed trades.

Combining the competitive nature and yearly records in profitability of this industry creates motivation for achieving ultra-low latency. One estimate puts the value of a millisecond of latency advantage at $100 million per year in arbitrage revenue [4], though latency alone is not the sole optimisable parameter in the trading ecosystem. Jitter is the measure of inconsistency or indeterminacy in the execution timing of a trading system, caused by unpredictable delays introduced by interrupts, buffering, and context-switching due to the Operating System (OS) kernel and networking stack. The dominating hardware for minimum jitter is the Field Programmable Gate Array (FPGA). Instead of executing instructions one at a time, an FPGA can be wired to perform a specific task in a single clock cycle with a fixed and predictable latency, regardless of how busy the system is, making it the preferred choice for HFT applications.

Research over the past two decades has steadily reduced this latency: from the sub-millisecond feed processing demonstrated by Thomas, and Luk [1], to the one-microsecond end-to-end routing achieved by Lockwood et al. [4], to the six-nanosecond single-cycle arbitration of Denholm et al. [8] Each of these represents a different layer of the same problem, and together they form a largely solved baseline.

This progress is approaching a hard physical boundary. The round-trip signal time along the Twinax copper cables of a co-location facility is already below $100ns$. No improvement in software, firmware, or chip design can make a signal travel faster than the wire allows. The next competitive frontier requires a functionally different kind of architectural thinking than a race for faster clocks.

Co-location imposes strict physical constraints beyond just speed. Exchange operators typically allocate each firm a fixed rack of servers drawing no more than $10kW$ of power, with no room to expand. Within a fixed power limit, a firm's strategy complexity is bounded by how efficiently its hardware converts power into useful computation.

This creates tension at the heart of today's HFT architecture. The synchronous FPGA designs that have produced the aforementioned latency results are fast because they are simple and predictable; every signal path through the chip is carefully designed to meet setup and hold constraints, a milestone dubbed timing closure. Superposing layers of intelligence to such an architecture, like neural networks capable of recognising scalable price patterns, requires adding logic, creating longer signal paths, henceforth breaking timing closure.

A metamodern class of hardware has emerged: event-driven many-core processors and neuromorphic chips, that deliver orders of magnitude more computation per watt than conventional approaches [5] [6]. Modelled after biological neurons, these systems process data through discrete spikes sent across a web of synaptic connections, only consuming energy when a specific input stimulus triggers a neuronal fire. The trading-aspect challenge is that they achieve this efficiency partly by abandoning the strict timing abidance that make synchronous FPGAs safe for financial executions. They are faster than conventional computers at inference, but slower than FPGAs at the nanosecond-scale order execution that HFT requires, and they introduce jitter through their non-

deterministic operation.

We examine hardware acceleration for low-latency financial computing. The field in question spans two communities that share a common objective (solving the von Neumann bottleneck) but rarely touch on each other's work. On one side, FPGA engineers build sub-microsecond data paths for high-frequency financial signal processing. On the other, neuromorphic designers build ultra-low-power spiking architectures for pattern recognition. Both communities claim that event-driven computation is the correct response to the limitations of conventional processors. Both measure success in orders-of-magnitude improvement. Neither has attempted a full system that bridges their domains.

This survey provides a cross-domain synthesis of these parallel efforts by reviewing nineteen papers published between 2008 and 2025 (Fig. 1), covering FPGA pipeline acceleration for trading [4] [8] [12], FPGA softcore cluster computing [1] [3], [11], neuromorphic chip architecture [6], [9], [17], [19], emerging-device fabrication [14], Spiking Neural Networks (SNN) for financial prediction [5], [10], [16], empirical flash-crash analysis [18], agent-based market simulation [15], FPGA deep-learning acceleration surveys [13], and software-based event processing for HFT [7]. We aim to map the field by identifying what is collectively believed, what it contests, and what it assumes without evidence by tracing three concept lineages (event-driven computing, FPGA acceleration, and spiking neural networks as computational primitives) through their origins and challenges, with their refinements over time. Through this analysis we flag five research questions that no paper in this corpus answers, then categorise every paper by evidence strength.

The FPGA literature forms the base of this field; Section II begins there. The architectural conviction uniting those FPGA papers with neuromorphic computing is event-driven execution, traced in Section III. Sections IV and V move from platform to application. Section IV asks what hardware should implement the event-driven paradigm, examining neuromorphic architectures whose fabrication have not been realised yet. Section V asks whether any of it can yield profits, evaluating the evidence for SNN-based trading. Section VI turns inward to how we built this collection, and the biases attached. Section VII identifies five gaps that no paper fills, and Section VIII concludes.

II. FPGA Acceleration for Financial Signal Processing

FPGA-based acceleration is the most tested part of this literature. Eight of the nineteen papers implement designs on real FPGA silicon with measured performance numbers. The field has bifurcated into two philosophies that solve very different problems.

A. Pipeline Architectures for Market Data

Thomas and Luk [1] produced the earliest work in this corpus, implementing a Monte Carlo credit-risk simulator on a Xilinx Virtex-4 at $233MHz$. The event-based model of loan defaults achieved 60–100$\times$ speedup over four parallel threads on a quad-core Pentium-4. The speedup range depended on

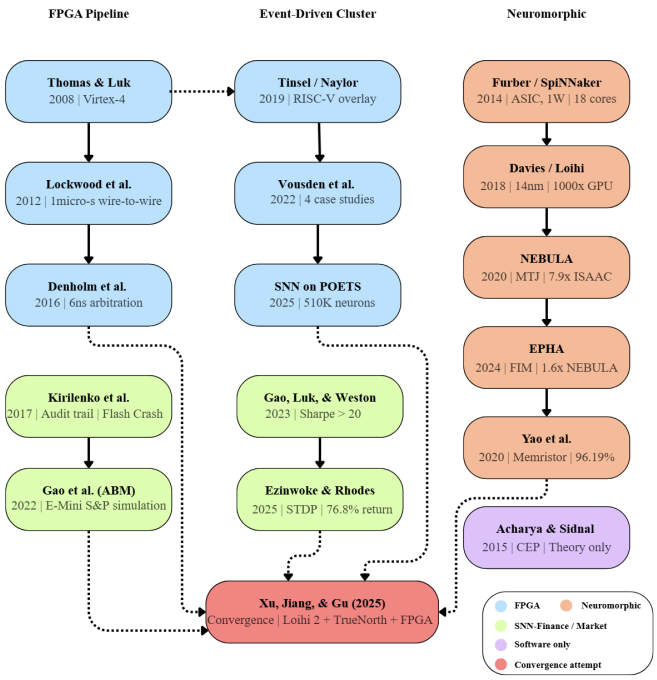


Fig. 1. Intellectual lineage across the papers in the corpus. Three parallel tracks (FPGA pipeline, softcore cluster, neuromorphic) converge at Xu et al. (2025).

input characteristics, a nuance ignored in most subsequent papers who instead reported single-point numbers.

Lockwood et al. [4] shifted the target to live trading infrastructure, developing an IP library on the NetFPGA-10G platform with parsers for FIX, OUCH, and ITCH protocols. Their application sustained 10 Gb/s line rate with $1\mu s$ fixed latency, two orders of magnitude below software implementations. The library addressed development time, a big practical barrier to FPGA adoption in finance since protocols, markets, and strategies evolve quickly over time. With secondary regard to performance, pre-verified IP blocks allowed trading firms to build custom logic without months of low-level FPGA system interconnect and input/output orchestration.

Denholm et al. [8] performed A/B feed arbitration on a Xilinx Virtex-6 and Virtex-5 FPGAs, and achieved single-cycle processing at $6ns$ for TotalView-ITCH and $5.25ns$ for OPRA and ARCA. They tested with real NASDAQ data from 9 September 2012, processing 322 million messages per training day. The design showed latencies one order of magnitude lower than a commercial Enyx system and $4.1\times$ lower than the IBM PowerEN processor. This remains the strongest methodology in the survey's entire collection of papers. Real FPGA and real market data measured in nanosecond latency. The high-reliability mode processed expected packets in 7 cycles ($42ns$ for TotalView-ITCH), and the low-latency mode could be output simultaneously.

Boutros et al [12] proved that High-Level Synthesis (HLS) can match hand-coded Register Transfer Logic (RTL) for complete trading systems. Their HFT pipeline built in C++ on

a Kintex Ultrascale XCKU115 at $156MHz$ achieved $869ns$ round-trip latency with network stack, FAST protocol parsing, as well as order book handling taken into account. This is within the same order of magnitude as Lockwood's RTL-based $1\mu s$ with faster development. The system used less than 8% of logic resources and zero Digital Signal Processing (DSP) blocks. This leaves the majority of FPGA compute available for adapting more advanced trading algorithms. Boutros also achieved $80ns$ average latency for order book insertion and deletion, which no prior work in this corpus had published.

B. Softcore Cluster Architectures

Parallel FPGAs have reconfigurable gates and host thousands of soft processors. Naylor et al. [11] built Tinsel, a hyperthreaded RISC-V overlay for DE5-Net FPGA clusters. Twelve FPGAs running $12,288$ threads demonstrated linear scaling on graph processing with an order-of-magnitude energy reduction over a Xeon cluster using the same power ($50W$ per DE5-Net board versus $100W$ per Xeon server). Tinsel trades-off single-thread performance for area efficiency and throughput, thereby making it the infrastructure layer for the POETS research programme.

Vousden et al. [2] built the event-driven paradigm on top of Tinsel, demonstrating a logarithmic scaling across four independent domains on the POETS platform (Dissipative Particle Dynamics (DPD), micromagnetics, Muller pipeline analysis, and genomics). In all four cases, event-based runtime grew sub-linearly contrary to traditional approaches which grew super-linearly, and the gap widened with problem size.

The SNN-on-POETS paper [3] extended POETS to brain-scale neural simulation. A uniform Synfire Ring SNN on six DE5-Net FPGAs achieved $509,648$ neurons and $509,648,000$ synapses in real time, representing 83.8% of the theoretical maximum predicted by an analytical model. Comparative analysis with SpiNNaker [19] showed POETS achieving $1,327$ neurons per core versus 512. This comparison carries a caveat, being that the POETS topology was optimised for POETS hardware. SpiNNaker numbers came from independent benchmarks under a different topology.

C. Limitations

Three factors weaken the FPGA evidence base. Firstly, no HFT pipeline paper reports power consumption, complicating comparisons with neuromorphic alternatives (Fig. 2); as long as the FPGA stays under the $10kW$ co-location limit, HFT engineers focus on tick-to-trade latency and jitter. Secondly, no paper tests a complete end-to-end trading system. Denholm measures the arbitration stage, Lockwood measures parsing, Boutros tests a thresholding strategy, but the decision-making unit that would connect these components is always assumed to exist downstream and unmeasured in these papers. Third, the latency-profit relationship that motivates the entire FPGA-for-HFT programme is assumed linear and never tested. Lockwood cites an industry claim that every millisecond of latency reduction improves arbitrage profitability by over $100M per year; no paper validates this. Xu et al. [5] later found evidence suggesting it may be wrong at the systemic level.

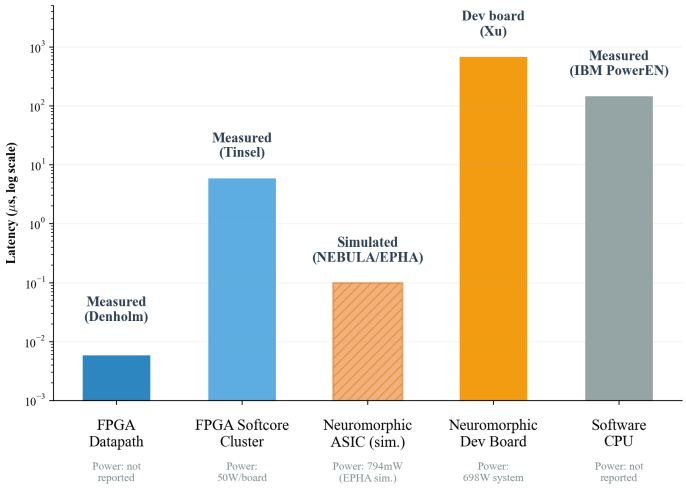


Fig. 2. Latency comparison (log scale) across substrates with power annotations. Hatched bar indicates simulated results. No substrate has both latency and power measured on the same financial task.

III. Event-Driven Computing

Event-driven computing connects FPGA simulation, softcore clusters, and neuromorphic spikes under the same umbrella. Computation fires only in response to stimuli instead of periodically fixed to a clock. A clocked processor burns power every cycle whether or not useful work is done, stalling whenever data must cross between processor and memory. An event-driven architecture operates mitigates that waste by consuming very little power in idle regions because no clock toggles their logic. Data moves as small messages between neighbouring processing elements (or, "devices"), bypassing the memory wall. Many cores can operate concurrently, so a much higher throughput rate can be expected. Eight of nineteen papers depend on this principle, and it is the strongest unifying idea in the collection.

A. The Paradigm

Thomas and Luk [1] used event-driven processing as an algorithmic technique for credit-risk modelling, treating loan defaults and environmental transitions as Poisson-process events. The fixed-timestep alternative would oversample quiescent loan classes and undersample active ones. Event-driven execution eliminated that waste.

Fourteen years later, Vousden et al. [2] elevated this technique to a general paradigm. Four case studies on the POETS platform all exhibited the same scaling advantage. Event-based runtime grew sub-linearly with problem size, while traditional big-compute approaches grew super-linearly. The gap widened as problems scaled. This is the broadest test of the event-driven thesis in the collection, and it holds across four domains governed by different physics.

The result builds on a sustained research programme. Thomas and Luk established event-driven financial simulation in 2008. Naylor et al. [11] built the RISC-V overlay (Tinsel) that gave POETS its hardware substrate. Vousden et al. proved the

paradigm generalises across domains. The SNN-on-POETS paper [3] extended it to brain-scale neural simulation. Each paper builds on the infrastructure and insights of its predecessors.

B. Software Based Approaches

Acharya and Sidnal [7] proposed a software-only Complex Event Processing (CEP) framework for HFT, their main argument being that stream processing suffers from a silo problem (detecting patterns within a single stream) while CEP can correlate events across heterogeneous streams. Lockwood et al. [4] and Denholm et al. [8] each process one market data feed in isolation. A trading firm watching various financial instruments simultaneously needs cross-stream pattern detection that CEP engines are specifically good at. Acharya describes an architecture with input adapters normalising heterogeneous feeds into a unified event stream with a pattern-matching engine correlating across them.

The difficulty is execution-related. CEP engines evaluate pattern definitions at runtime through an interpreted matching process using a mathematical model called a Non-deterministic Finite Automaton (NFA), for each, an incoming event is tested against every open detection window. Each unresolved partial match creates an active "pattern-matching automaton" instance. This is manageable under normal market conditions, but during a volatility spike, event arrival rates can abruptly increase by two orders of magnitude. The active-instance count grows combinatorially with event rate and window duration. CEP engines cannot bear the stress of this growth or data load at the same speed as the information incoming from the stock exchange during an event-storm, per se. One can argue that the moments cross-stream correlation matters most (earnings surprises, flah crashes, etc.) are the moments when the pattern-matching automaton is least able to keep up.

C. Non-Stationary Domains

Event-driven computation on POETS is already non-stationary at the hardware level. The NoC delivers packets in non-deterministic order. Routing is asynchronous and path-dependent, so no core can predict when its next message will arrive. Different regions of the mesh exhibit different activity levels at different times as physical hotspots migrate across the problem space, meaning the boundary between active and quiescent regions shifts continuously. The event choreography, stochastic from the perspective of any individual core, is stiff from the perspective of the global mesh. POETS can handle this by letting threads sleep when their region is quiet and fire when stimulated; the concept of adaptive load distribution is the cause behind the sub-linear scaling Vousden et al. [2] demonstrated.

Financial markets introduce a second layer of non-stationarity. In physics simulations, the compute pattern shifts but the governing equations do not. DPD particles obey the same conservation laws throughout the run. Micromagnetic dynamics follow the Landau-Lifshitz-Gilbert equation regardless of measurement. The rules generating events are fixed even when the spatial distribution of these events is not.

Re-iterating, financial markets present the challenge that they are reflexive. The act of trading on a signal changes the signal. Event statistics shift through unknown events too niche or too contemporarily unpredictable, pushing the HFT industry away from the use of Application-Specific Integrated Circuits (ASICs) that would become outdated in the next economic transformation; firms favor the simpler maintenance of deploying reprogrammable FPGAs instead. Whether event-driven architectures can maintain their scaling advantage in domains where the dynamics are non-stationary is an open and important research question. No paper in this collection tests it at the platform level. Answering it would extend the event-driven paradigm into a demanding application domain and represents a plausible next step for the research programme that Vousden et al. established.

With the event-driven computing paradigm established, the topic to follow is the hardware that should (or can) implement it. FPGAs emulate event-driven behaviour through software overlays on clocked silicon. A different class of architecture builds event-driven operation into the device itself.

IV. Neuromorphic Architectures for Inference

Four architectures compete in this literature. Two of which are fabricated, and the remainder exist only in simulation.

A. Fabricated Chips

Furber et al. [19] designed SpiNNaker, a million-core computer with $200MHz$ ARM968 processors consuming $\approx 1W$ per 18-core chip, each chip containing a packet router that routes neural spike packets as per their source identifiers through a 2D triangular mesh using multicast transmission. The architecture supports real-time simulation of large-scale SNNs and is the most-cited platform in this collection, referenced by six papers.

Davies et al. [17] designed Loihi, a $60mm^2$ chip fabricated in Intel's $14nm$ process. Loihi integrates 128 neuromorphic cores, each containing $1,024$ spiking neural units, with programmable synaptic learning rules including Spike-Timing-Dependent Plasticity (STDP) variants. Running a spiking convolutional form of the Locally Competitive Algorithm (LCA), Loihi solved LASSO optimisation problems with over three orders of magnitude better energy-delay product than conventional CPU solvers at matched process, voltage, and area. [17] finds that spike-based computation outperforms all known conventional approaches on a specific problem class. The result generalises to recurrent networks, which makes it significant beyond the LASSO benchmark.

Yao et al. [14] built a five-layer memristor Convolutional Neural Network (CNN) integrating eight $2,048$-cell crossbar arrays fabricated in a TiN/TaO_x/HfO_x/TiN material stack. Using a hybrid-training scheme that adapted to device imperfections, the system achieved 96.19% MNIST accuracy and $110\times$ higher energy efficiency (11.014 GOPS/W) than the NVIDIA Tesla V100 GPU. This is the only fabricated emerging-device system in the collection. It is also the only paper where a competing device class (memristors, as opposed to the spintronics used in NEBULA [9] and EPHA [6]) has crossed the fabrication

threshold. The crossbar arrays exhibited uniform analogue switching under identical programming condition, a property that NEBULA and EPHA assume but cannot verify without silicon.

B. Simulated Architectures

Singh et al. [9] proposed NEBULA, a spin-based architecture using Magnetic Tunnel Junction (MTJ) devices. On standard benchmarks (AlexNET, VGGNet, MNIST, CIFAR-10, ImageNet), NEBULA achieved up to $7.9\times$ energy efficiency over ISAAC in ANN mode and approximately $45\times$ over the INXS SNN accelerator. NEBULA used dedicated incompatible neural cores for ANN and SNN processing. When one core type was active, the other sat idle, wasting hardware.

Zhao et al. [6] proposed EPHA using compensated ferrimagnet (CoGd-FIM) devices. A unified neural core eliminated NEBULA's core-idleness problem by switching between multi-bit analogue drivers (ANN) and binary spike drivers (SNN) on the same crossbar. EPHA claimed $1.6\times$ better power efficiency in ANN mode, $27\times$ computational efficiency, $16\times$ in SNN mode, and four orders of magnitude over Loihi [17]. The Loihi comparison deserves careful reading, as EPHA's numbers come from simulated ferrimagnet device models. Loihi's numbers come from measured $14nm$ silicon; a different category of evidence. EPHA itself acknowledges that fabrication variation remains unresolved, which means the simulated advantage may narrow once real devices replace ideal ones.

C. The Fabrication Gap

Neither NEBULA nor EPHA has been fabricated. Efficiency results are simulated using mumax3 device-physics models and analytical frameworks, like the Bitlet model.

On one hand, silicon has had sixty years of industrial process development. The full fabrication process has been refined across a plethora of foundry generations. The automation of electronic design toolchain assumes silicon.

On the other, non-silicon neuromorphic devices face the issue that the materials making them computationally desireable (for spintronics) depend on atomic-scale properties that simulation tools model as ideal but fabrication cannot guarantee. An MTJ's conductance, for instance, depends on the thickness of an oxide barrier measured in individual atomic layers. A memristor's resistance states depend on the geometry of its conductive filament, stochastic by nature. The gap between simulated and fabricated behaviour comes from building components from materials whose function depends on details no lithographic process fully controls, presently. For an edification of these challenges in the spintronic case, we refer the reader to Grollier et al. (2020).

Yao et al. [14] deemed this gap crossable. Their memristor CNN achieved 96.19% accuracy on fabricated crossbar arrays that exhibited real device variation, sneak-path currents, and yield imperfections. The system used a hybrid-training scheme adapted to the imperfections measured on real silicon. This is a potential path forward for all non-silicon neuromorphic computing. Neither NEBULA or EPHA has attempted it because neither has silicon to adapt to.

The Shawahna et al. [13] survey of over fifty FPGA DNN accelerators adds a further competitive pressure. FPGA platforms already achieve moderate-to-high inference performance at lower power than GPUs. The non-silicon papers must outperform GPUs and a maturing FPGA accelerator ecosystem to perhaps justify investment toward fabrication.

These architectures were designed for neural simulation and image classification. Their profitability in live markets is a separate question.

V. SNN Approaches to Financial Trading

Three papers bridge neuromorphic computing and finance. A fourth discusses the systemic risks they raise, and a fifth provides the simulatory framework needed to study those risks.

A. STDP for Finance

Gao, Luk, and Weston [16] originated the thesis that STDP-trained SNNs can predict price spikes in financial time series. They used intraday tick data for crude oil and gold futures, demonstrating that spiking networks learn hidden temporal patterns through unsupervised STDP. Three trading strategies combining spike-learning with traditional approaches produced Sharpe ratios uniformly above 20 in backtesting, a result that far exceeds the most profitable systemic strategy in history. One issue with this metric is the data; the backtest covers only 23 trading days whereas a Sharpe ratio is an annualised metric. If the hyperparameters were chosen and tuned with awareness of performance on the 23-day dataset, then this would suggest an overfitting of results, but the paper does not make that presumption clear. A transfer experiment (training on crude oil futures, testing on gold futures) showed no significant performance degradation, suggesting the learned patterns are cross-asset, not necessarily cross-class; the models were trained and tested on commodities which historically, are highly linked. The paper explicitly connected SNN pattern recognition to technical analysis, arguing the network performs an automated, higher-frequency version of chart-pattern detection. The computational performance limitation that Gao identified as future work (meeting microsecond-level latency requirements) would require the neuromorphic hardware described by Davies [17] and Furber [19].

Ezinwoke and Rhodes [10] extended and challenged Gao's work. They trained three SNN architectures on real AAPL tick data (microsecond precision, over 19 trading days in February 2015). Model 1 replicated Gao's STDP approach. Model 2 added explicit inhibitory synapses and enriched temporal inputs. Model 3 used supervised backpropagation. The paper introduced Penalised Spike Accuracy (PSA), a novel Bayesian Optimisation (BO) objective ensuring the predicted spike aligns with the empirical rate of price events. This addressed a vulnerability in the original Spike Accuracy metric which rewarded networks that suppressed spiking in an attempt to reduce energy consumption.

Model 2 with PSA achieved 76.8% cumulative return, a Sharpe of 19.71, and 53.49% win rate. The supervised baseline achieved 42.5% return with a Sharpe of 15.91. These results are consistent with Gao's findings across a different asset class and

time period, strengthening the thesis for STDP applications to finance. Nineteen days of one stock is not enough to generalise. Furthermore, no transaction costs beyond spread were modelled, and the models ran in software rather than on neuromorphic hardware.

B. Neuromorphic System Architecture

Xu, Jiang, and Gu [5] designed a heterogeneous system integrating Loihi 2 (development version), TrueNorth, SpiNNaker 2, Xilinx Alveo FPGAs, and NVIDIA A100 GPUs. Four layers handled perception, computation, execution, and risk monitoring. The paper claimed 78.3% latency reduction over traditional architectures, 80% energy savings, and a $3.2\times$ speed increase over an FPGA/GPU hybrid cluster. An ablation study showed disabling the event-driven mechanism caused 67% throughput loss. Replacing the neuromorphic processor with a GPU reduced energy efficiency by 81%.

The reproducibility of this methodology is difficult to assume due to the "traditional architecture" baseline they measured never being defined. The paper's claim that "92.1% reduction in energy consumption on FPGA-specific chips" appeared without enough detail for independent verification. The system ran on development boards, not production silicon.

C. Market-Stability Analyses

Xu et al. [5] found that algorithmic trading exceeding roughly 65% of total volume triggers a feedback coefficient surge of 0.57 and flash-crash probability of 76.2% using five years of Chinese A-share data (CSI 300, 2019-2024). This challenges the entire acceleration literature.

Kirilenko et al. [18] analysed the 2010 Flash Crash using CFTC audit-trail transaction-level data for the E-Mini S&P 500. They classified traders into six categories and tracked their behaviour before and during the crash. High-frequency traders did not change their trading patterns when prices fell. They did not trigger the crash, they bought during the initial decline and then quickly sold. This in turn exacerbated volatility and accelerated the price drop. The crash originated from a large institutional sell order of $75,000$ E-Mini contracts (worth $\$4.1$ billion) executed by an algorithm that targeted 9% of trading volume without regard to price or time.

These findings are complimentary. Kirilenko analyses one catastrophic event with the highest-quality financial data available (regulatory audit trails) and finds HFTs amplified but did not originate the crash. Xu analyses five years of aggregate data and finds a statistical threshold for destabilisation. Appropriately, one may ask what happens when different species of hardware compete in one order book.

Gao et al. [15] built the framework to answer that question. Their high-frequency agent-based model of the E-Mini S&P 500 reproduced the 2010 Flash Crash at $100ms$ resolution, calibrated using machine learning surrogate modelling. Monte Carlo simulations identified three conditions that significantly affect crash amplitude: (i) the percentage of volume executed by the Sell Algorithm, (ii) market maker inventory limits, (iii) the trading frequency of high-impact traders. The relationship between crash amplitude and sell-algorithm volume is non-monotonic, a finding that complicates the simple narrative of more speed equals more crashes. The model is validated against twenty real-market stylised facts using statistical tests and moment coverage ratios, and provides the infrastructure for studying hardware-heterogeneous agent populations. That experiment has not been attempted in this survey's corpus.

The claims in the preceding sections vary in the weight of evidence behind them. The next section examines how we assembled this collection and what each category of evidence can and cannot establish.

VI. Research Methodology and Evidence Evaluation

A. Paper Selection Methodology

This survey was constructed through targeted searches across IEEE Xplore, ACM Digital Library, Nature, Scopus, and Google Scholar using combinations of terms including (but not limited to) "FPGA high-frequency trading", "neuromorphic computing finance", "spiking neural networks trading", "event-driven hardware acceleration", and "flash crash algorithmic trading". We supplemented keyword searches by following citation chains. Papers that appeared in the reference lists of multiple collected works were treated as candidates for inclusion. This is how the foundational papers (SpiNNaker [19], Loihi [17]) and the flash-crash literature (Kirilenko et al. [18]) entered the collection.

The resulting corpus has identifiable biases. First, it majorly represents the Imperial College London and University of Southampton research groups. Seven of the nineteen papers trace to these institutions. This occurred because the FPGA-for-finance literature is a niche of the mainstream FPGA corpus and this group has been among the most productive contributors to this niche. The bias is a reflection of the field's concentration and not a deliberate selection choice, but it means the survey gives disproportionate weight to one intellectual lineage. Second, the corpus favours measured FPGA experiments over simulation-only neuromorphic work. Eight papers report results on real FPGA silicon, two report results on fabricated neuromorphic chips, and two rely entirely on device-level simulation. This distribution reflects our preference for field-tested evidence over analytical models when selecting papers for inclusion. It does mean the neuromorphic-architecture debate (NEBULA versus EPHA) is represented by fewer papers than a dedicated neuromorphic survey would include. Third, the financial-economics side of the literature is underweighted. Two papers (Kirilenko et al. [18] and Gao et al. [15]) address market microstructure. A dedicated survey of HFT market impact would include Budish, Cramton, and Shim (2015) on the latency arms race, Menkveld (2013) on HFT market-making, and Brogaard, Hendershott, and Riordan (2014) on HFT price discovery. We included the two papers most directly relevant to hardware-induced systemic risk and excluded the broader market-microstructure literature to maintain the survey's focus on hardware acceleration.

Despite these biases, the collection captures the key results at each point where the three tributaries (FPGA acceleration, neuromorphic computing, and financial application) intersect. The SNN-for-finance niche is small; Gao et al. [16] and Ezinwoke and Rhodes [10] are the only two papers applying STDP to high-frequency price prediction with real tick data. The FPGA-for-HFT pipeline literature is also small; Denholm et al [8], Lockwood et al. [4], and Boutros et al. [12] are the most-cited implementations. The cross-domain intersection that this survey maps is hard to come across, in part because so few papers occupy it.

B. Evidence Critique Across the Collection

The nineteen papers produce five types of evidence, with independent analytical reach.

One type of evidence contains papers that combine real hardware with real-world data. Denholm et al. [8] tested on real NASDAQ feeds with measured nanosecond latency. Kirilenko et al. [18] used CFTC audit-trail transaction data, the highest-quality financial market evidence available to academic researchers. Davies et al. [17] and Furber et al. [19] report measured results from fabricated silicon (Loihi at $14nm$, SpiNNaker as a custom ASIC). Yao et al. [14] measured a fabricated memristor CNN achieving 96.19% accuracy and $110\times$ GPU energy efficiency.

Another contains FPGA experiments on synthetic or standard benchmarks. Thomas and Luk [1], Lockwood et al. [4], Boutros et al. [12], Naylor et al. [11], Vousden et al. [2], and the SNN-on-POETS paper [3] all report measured hardware results, but on synthetic data, standard graph benchmarks, or domain-specific simulation inputs.

A third type of evidence comes from computational experiments validated through backtesting. Gao et al. [16] and Ezinwoke and Rhodes [10] both use real financial tick data, but run their SNN models in software and validate through simulated trading. Gao et al. [15] calibrate an agent-based model against real stylised facts, a stronger form of validation than uncalibrated simulation, but still operating entirely in software.

Papers that rely on device-physics simulation take up another category, like Xu et al. [5] (development boards with undefined baselines), NEBULA [9] and EPHA [6] (device-physics simulation with no fabricated silicon) and Shawahna et al. [13] (a meta-level review contributing with design insights).

Finally, Acharya and Sidnal [7] who proposed a theoretical software-based CEP framework without implementation or experimental validation, forming a separate category of evidence based on contribution to the field's problem definition.

C. Absent Methodologies

Three methodologies remain entirely absent from the collection and, as far as our research indicates, from the field itself. No cross-substrate power benchmark measures the same financial workload on FPGA, neuromorphic ASIC, and GPU with real power draw. No production deployment study exists. Every paper operates in a laboratory or simulation environment, Firms that deploy FPGAs for trading do not publish. No longitudinal financial validation runs a model forward in time and measures performance against reality. Financial time series are non-stationary; a system that works in February 2015 may fail in March.

VII. Research Agenda

Five gaps are drawn from this research, Fig. 3 maps them against the paper clusters that define them.

A. Cross-Substrate Benchmarking with Measured Power

The FPGA papers never report power. The neuromorphic papers report simulated power. Xu claims 80% energy savings over an unspecified baseline. Yao [14] reports $110\times$ efficiency over a V100 GPU, the only measured emerging-device number in the collection. A controlled benchmark running NASDAQ TotalView-ITCH feed processing across FPGA datapath, FPGA softcore cluster, Loihi and GPU would settle debates that currently rely on incompatible measurement approaches. The Southampton and Imperial groups already have access to three of the four substrates.

B. SNN Trading Validation at Scale

Ezinwoke's STDP result relies on 19 days of one equity. Gao's Sharpe ratios above 20 come from crude oil and gold futures backtests with no transaction costs. Running both methods on at least three years of tick data across multiple asset classes, with realistic market impact and slippage models, may determine whether the STDP advantage survives regime changes. The critical test is concept drift: if STDP is really better at non-stationary temporal structure, it should outperform supervised methods precisely during market transitions.

C. Event-Driven Financial Simulation on POETS

Thomas and Luk proved event-based credit-risk simulation works on a single FPGA in 2008 [1], then POETS proved event-driven computing scales sub-linearly in 2022 [2]. Thomas co-authored both, yet nobody has ported the 2008 financial model to the 2022 platform. This gap requires no new hardware and no new algorithms, it is a matter of implementation.

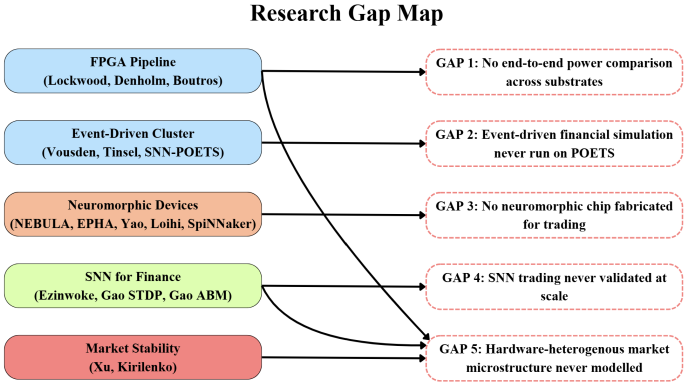


Fig. 3. Five open research questions, Gaps 2 and 4 are actionable with existing infrastructure, Gap 5 requires hardware fabrication.

D. Market Microstructure with Hardware-Heterogeneous Agents

Gao et al. [15] built a calibrated flash-crash simulator. Kirilenko et al. [18] found veritable evidence for the 2010 Flash Crash. Xu et al. [5] found the 65% algorithmic-trading threshold. All three treat algorithmic traders as equivalent, so an extension of Gao's ABM where CPU-speed, FPGA-speed, and neuromorphic-speed agents compete in one order book would be interesting to test whether hardware heterogeneity creates qualitatively different market dynamics. It would be interesting to compare neuromorphic-agent activity that uses spike-encoded order books in contrast to FPGA pipeline agents. The simulation infrastructure exists, but the experiment does not.

E. Fabrication of Spintronic Neuromorphic Chips

NEBULA [9] and EPHA [6] have competed in simulation for four years. Yao et al. [14] demonstrated that the competing memristor device class works on real silicon despite the fabrication variation both spintronic papers flagged as unresolved. The spintronic subfield will remain provisional until at least one design is taped out and measured at $28nm$ or below.

VIII. Conclusion

The field has established beyond reasonable doubt that purpose-built hardware processes financial data and runs neuromorphic computations hundreds of times faster and cheaper than conventional processors. Denholm's $6ns$ FPGA arbitration on real NASDAQ feeds [8], Loihi's three-orders-of-magnitude advantage on LASSO [17], Yao's fabricated memristor CNN at $110\times$ GPU efficiency [14], and Vousden's sub-linear event-driven scaling across four domains [2] are all measured on real systems.

The remainder of the collection's claims rest on analytical models, development boards, or datasets too narrow to generalise from. The neuromorphic-for-trading thesis has exactly two verifiable data points (Gao [16] and Ezinwoke [10]), both showing STDP outperforms supervised learning, neither tested on neuromorphic hardware or at production scale.

Trading systems are approaching a point where they may differ from existing ones in kind, not degree. If so, they will process prices through spike encodings, learn through biological synapse mechanisms, and make decisions through processes their designers cannot yet fully inspect. Kirilenko et al. [18] showed that even homogeneous algorithmic traders can destabilise a market in minutes. Xu et al. [5] found the threshold at which destabilisation becomes probable. Gao et al. [15] built the simulator to study it. What happens when architecturally alien systems enter a market already dominated by FPGA pipelines is a question that, in retrospect, will determine whether that future benefits or chaotises HFT firms' market operations.

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